Introduction

Machine Learning 1 — Lecture 1

10^{th} March 2026

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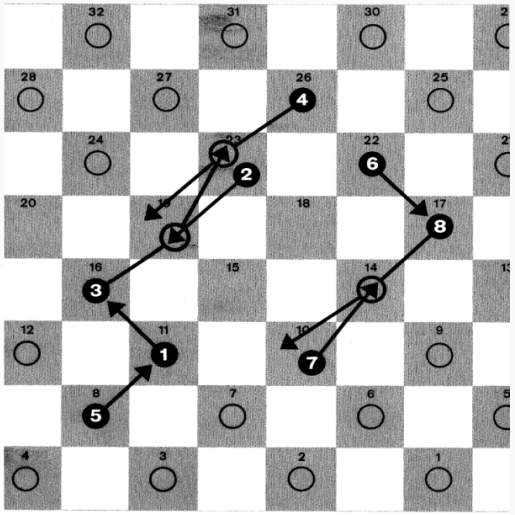
Arthur L. Samuel, 1959

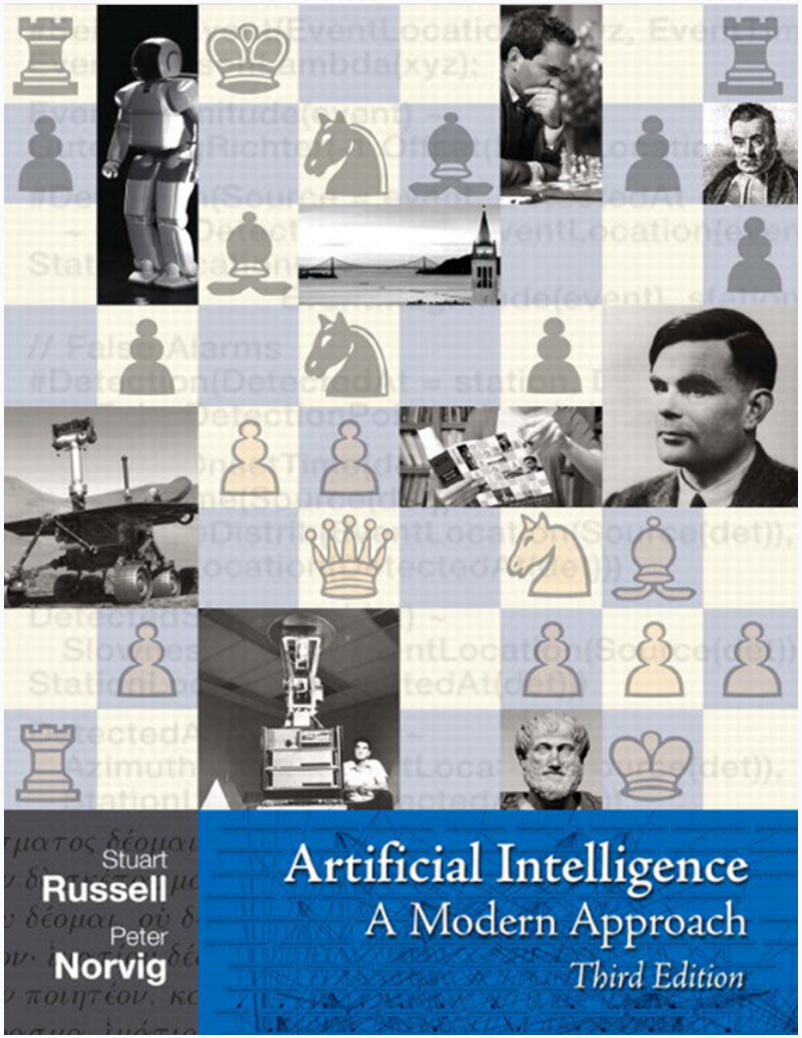


Eight-move opening utilizing generalization learning. (See Appendix B, Game G-43.)

Some Studies in Machine Learning Using the Game of Checkers

Abstract: Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these experiments are, of course, applicable to many other situations.

Machine Learning: Subfield of Artificial Intelligence



Four Historical Approaches to AI

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)
"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998)
"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Learning is any process by which a system improves performance from experience. — *Herbert Simon*

Learning is any process by which a system improves performance from experience. *Herbert Simon*

- system: (internal state of) animal/human/learning machine
- process: adaptation, change, optimization, algorithm
- experience: input, data, observations
- performance: loss, fitness, score, utility, objective

Example

"Write a program which distinguishes apples from pears."

Example

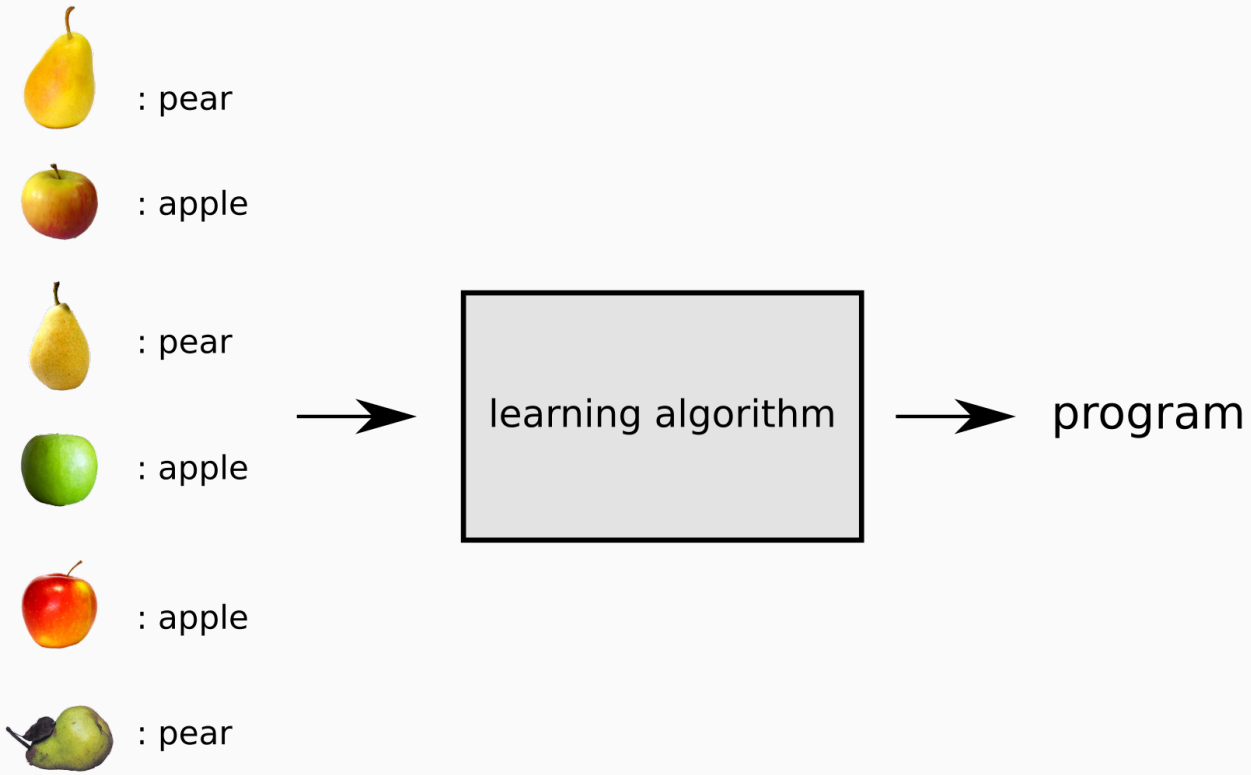
"Write a program which distinguishes apples from pears."

How to even start thinking about this task?

Example

It might be easier to:

- write a **meta-program**/**learning algorithm**, which is
- provided with **training data** (examples of input-output pairs)
- and which returns a program (or **model**) solving the task



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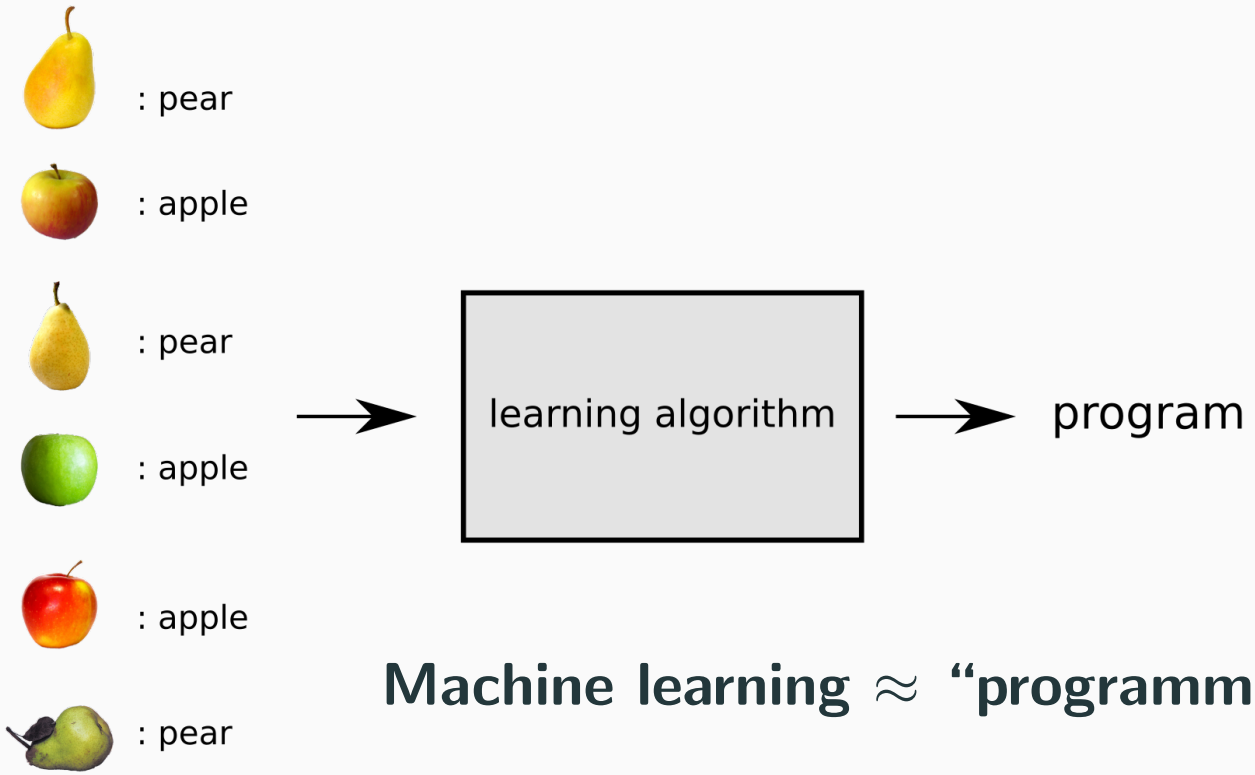
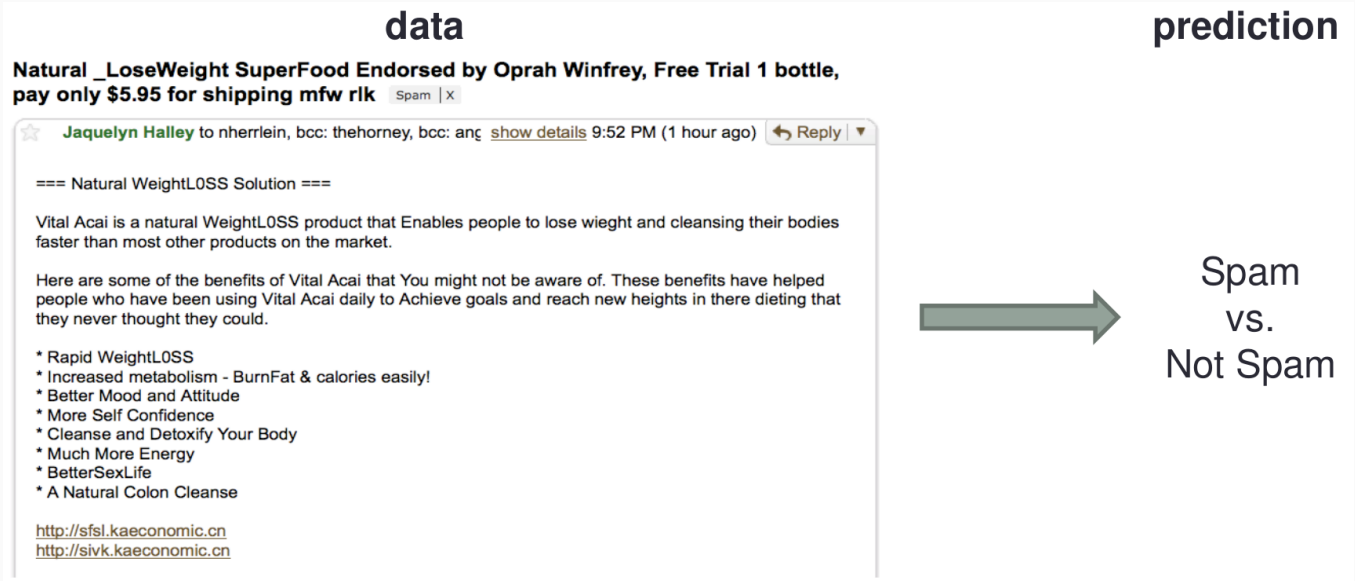


Machine learning $\approx$ "programming with data"

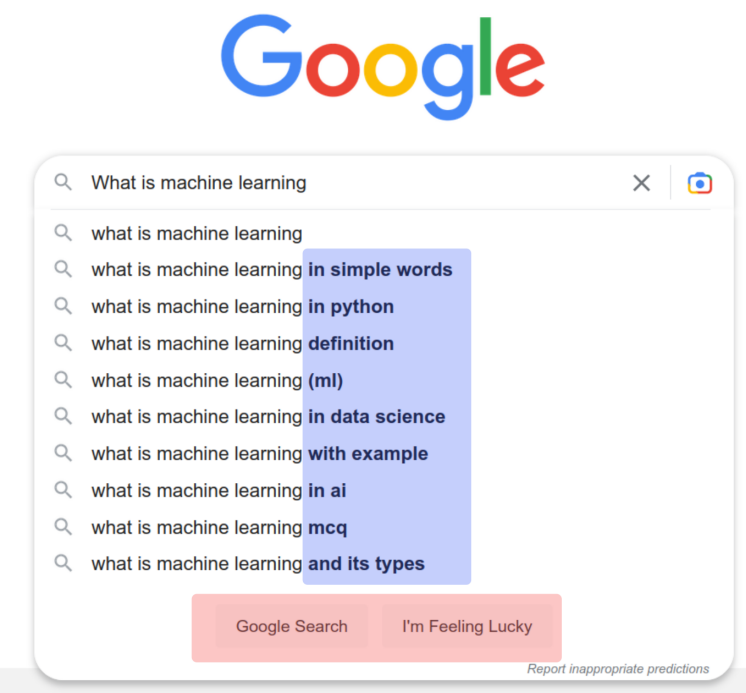
Spam Filtering

Spam in email started to become a problem when the Internet was opened up to the general public in the mid-1990s. It grew exponentially over the following years, and today composes some 80 to 85% of all the email in the world, by a "conservative estimate"

`https://en.wikipedia.org/wiki/Spamming`



Web Applications

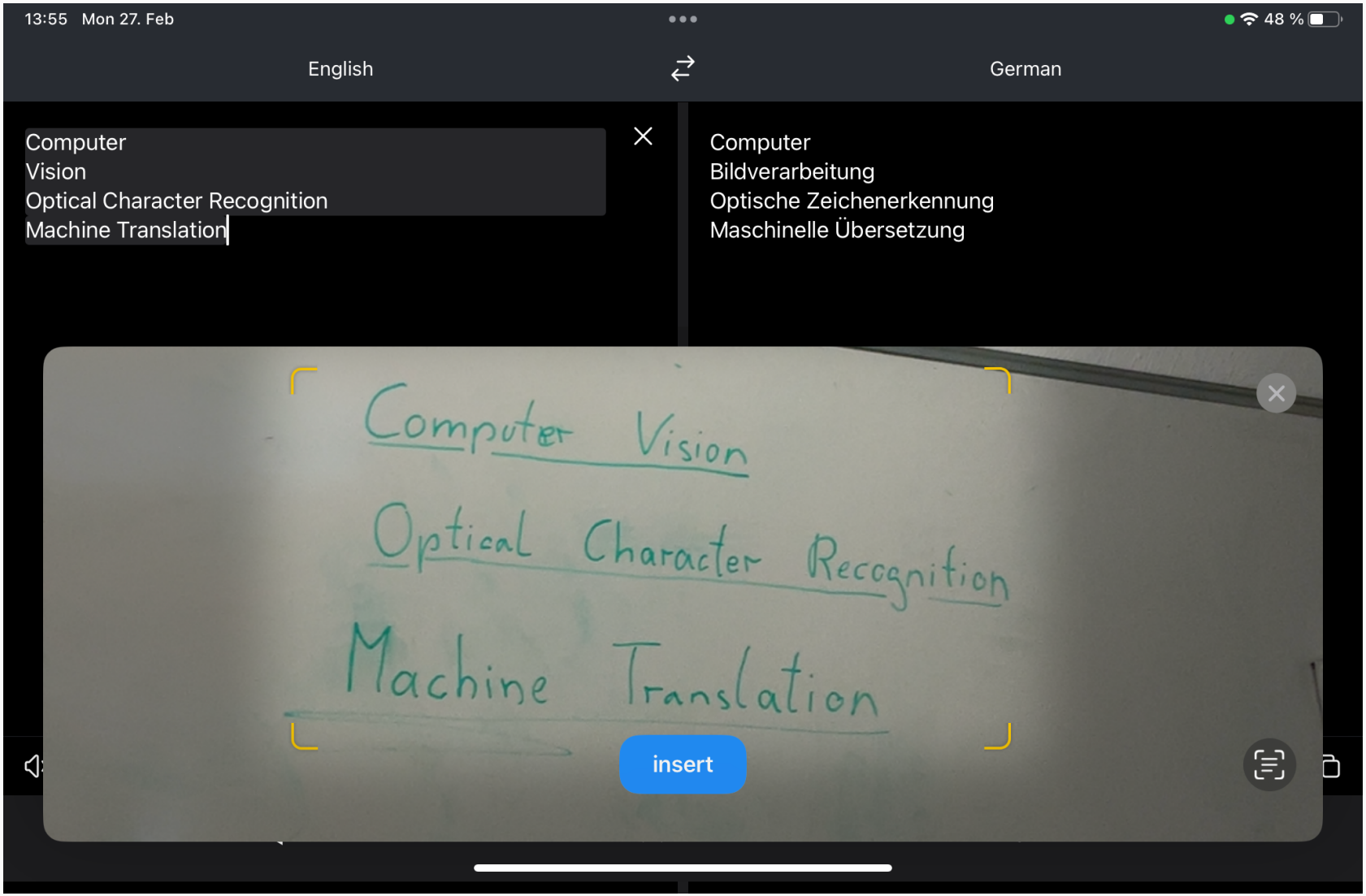


Web Search

Natural Language Processing

Advertising & Recommender Systems

Computer Vision, Machine Translation



Chat GPT

How many atoms are there in the Harry Potter Universe?

The number of atoms in the **Harry Potter Universe** depends on how we define that universe. Here are a few ways to think about it:

1. **If the Harry Potter Universe = Our Universe (Including the Wizarding World)**
   - The observable universe has approximately 10^{80} **atoms**.
   - If we assume the Harry Potter Universe exists within our own, the number of atoms remains the same.
2. **If the Harry Potter Universe = Just Earth (Where Most of the Story Takes Place)**
   - Earth has about 1.33×10^{50} **atoms**.
   - This includes all wizards, witches, Muggles, and magical creatures.
3. **If the Harry Potter Universe = Just the Books, Movies, and Fandom Materials**
   - A single printed book has around 10^{25} **atoms**.
   - If we count all Harry Potter books printed (~500 million copies), that's about 5×10^{34} **atoms**.
   - If we add movie reels, posters, wands, and merch, the number goes up but still nowhere near planetary levels.
4. **If the Harry Potter Universe = Hogwarts and Key Magical Locations**
   - Hogwarts Castle (~1 million tons of stone) contains about 10^{32} **atoms**.

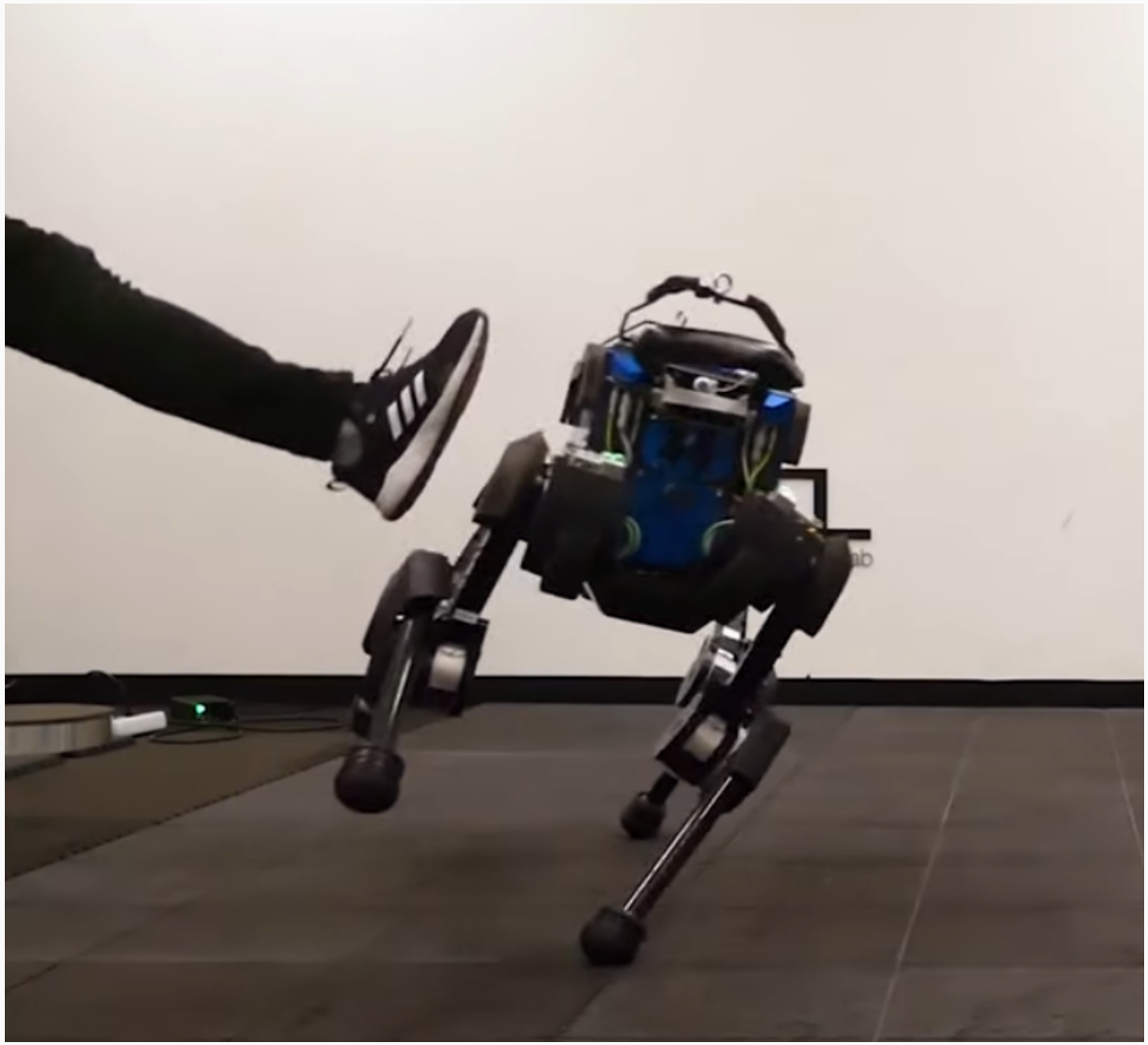
Self-driving Cars

https://www.youtube.com/watch?v=TsaES--OTzM

- many states have partially legalized self-driving cars (e.g. for testing)
- full autonomous driving (SAE level 5, "steering wheel optional")
  still a long way down the road

Robotics

https://www.youtube.com/watch?v=3jDoPobFgwA



Game Playing (Chess, Go, Video Games)

https://www.youtube.com/watch?v=7L2sUGcOgh0

Bioinformatics, Protein Folding

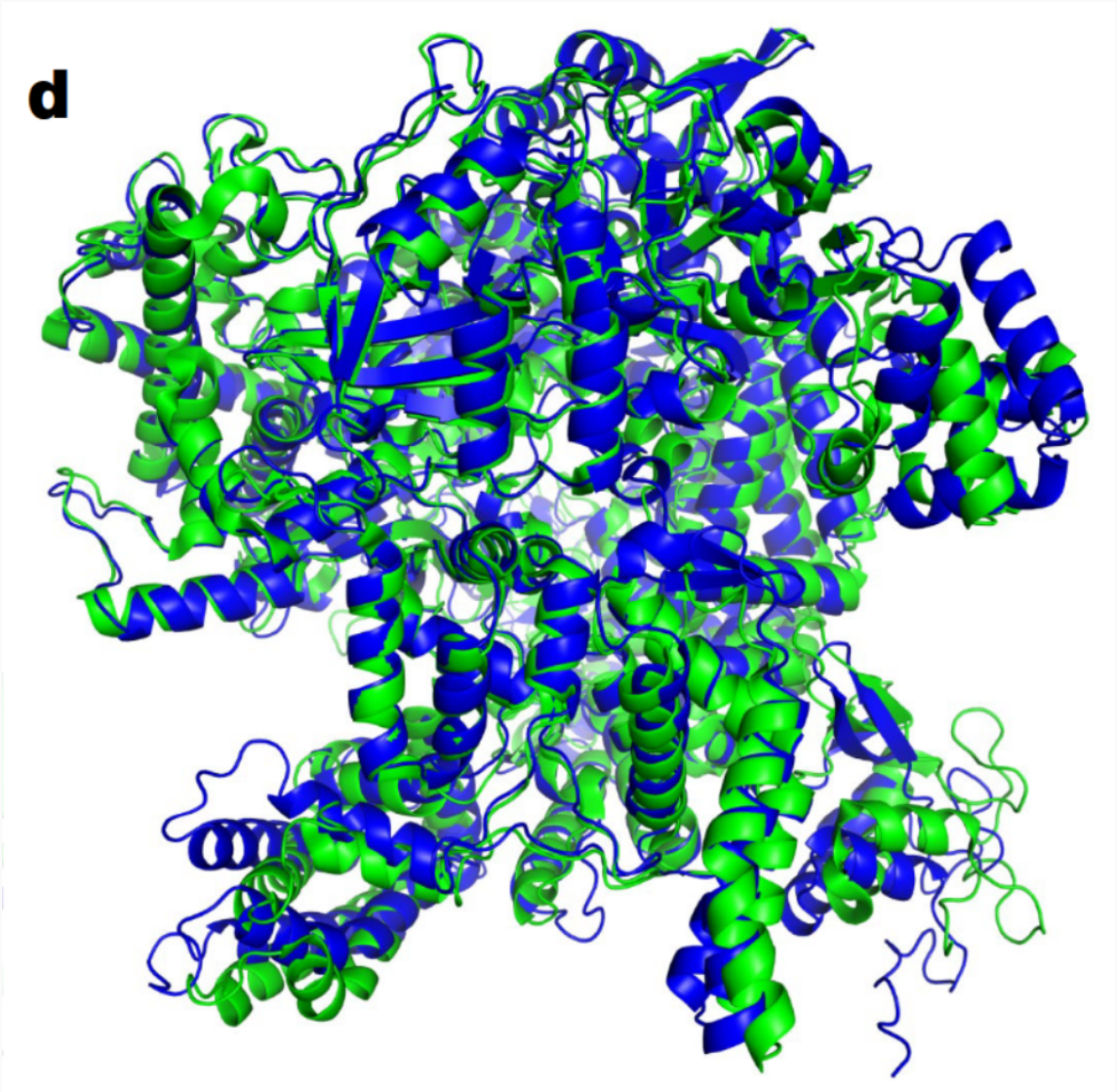
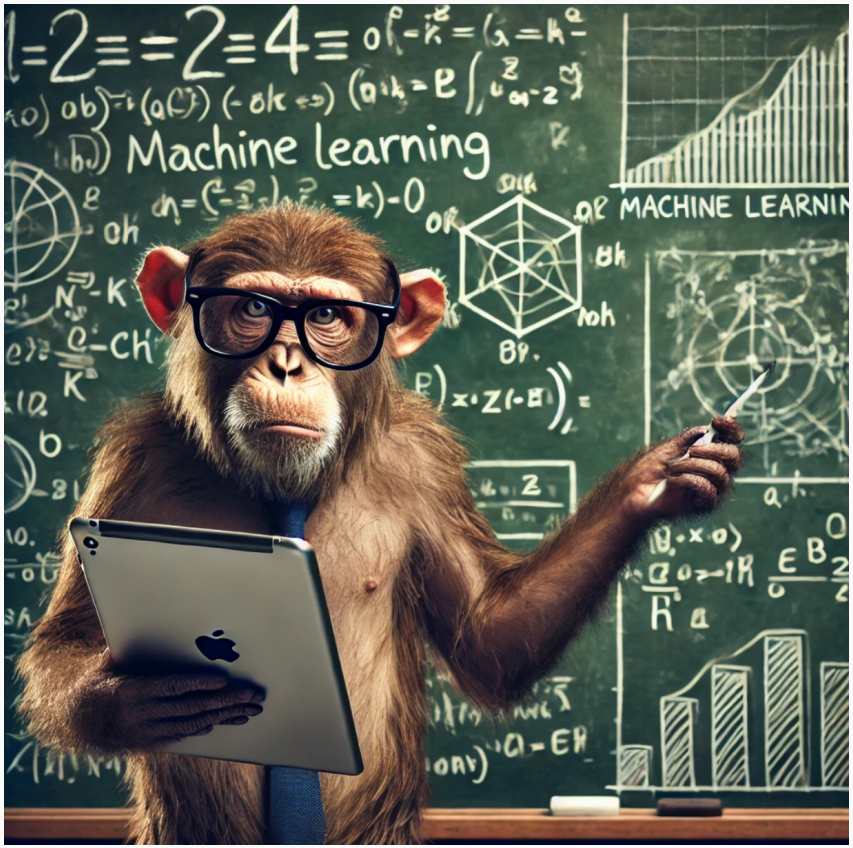


Image: Jumper et al., Highly accurate protein structure prediction with AlphaFold, Nature 2021

Image Generation

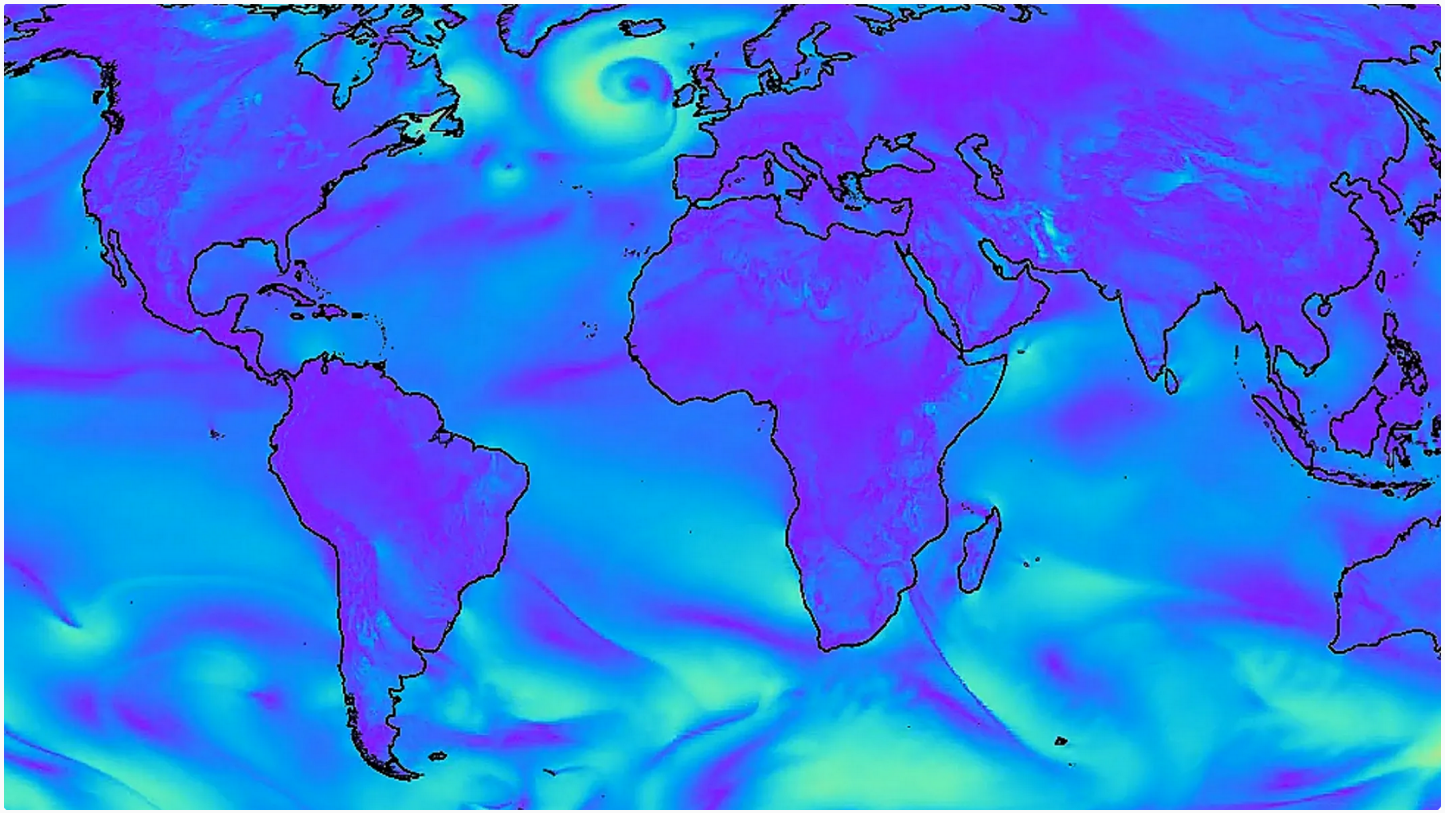
"Monkey with glasses and IPad, desparately trying to teach Machine Learning in front of black board with fancy math on it"



Generated by DALL-E, interfaced via ChatGPT

Weather Prediction

Weather prediction systems based on machine learning outperform classical numerical methods by now.



`https://deepmind.google/`

Applications

- astronomy
- autonomous driving/navigation
- bio-informatics
- brain-machine interfaces
- chemistry
- computational finance
- data visualization
- drug discovery
- event detection
- fraud detection
- game playing
- image generation
- information retrieval
- medical diagnosis (support)
- natural language processing
- network analysis
- online advertising
- personal assistants
- recommender systems
- search engines
- speech recognition
- speech synthesis
- speech and language translation
- visual object detection
- ...

Overview

Organizational Matters

Machine Learning Paradigms

Organizational Matters

What will you get from this course?

- a general and broad introduction to machine learning
- foundational and mathematical principles of machine learning
- theoretical understanding of most important machine learning algorithms
- hands-on experience in implementing and applying algorithms

Sessions, Lectures and Practicals

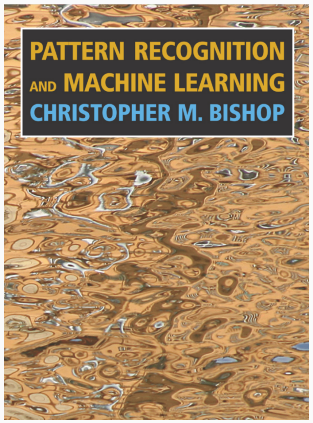
- **lectures are on Tuesdays, 11:30–13:15, HS i13**
- covers theory, principles and learning algorithms
- 2×45 minutes, ~ 10 minutes break

- **practicals are on Wednesdays, 10:30-11:30, HS i13**
- discusses applications, implementation and code

- in-person sessions, but no mandatory attendance—**except for the final exam! (29. and 30. of June)**
- all sessions will be recorded

Materials

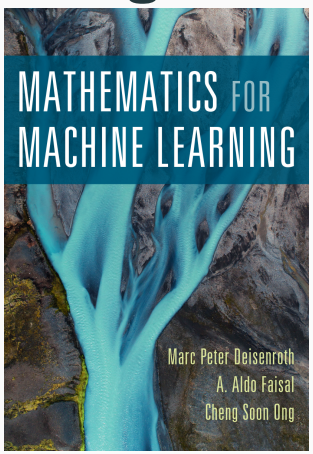
- lecture slides on TeachCenter
- Bishop, **Pattern Recognition and Machine Learning**



Freely available under:

`https://www.microsoft.com/en-us/research/people/cmbishop/prml-book/`

- Deisenroth et al., **Mathematics for Machine Learning**



Freely available under:

`https://mml-book.github.io/`

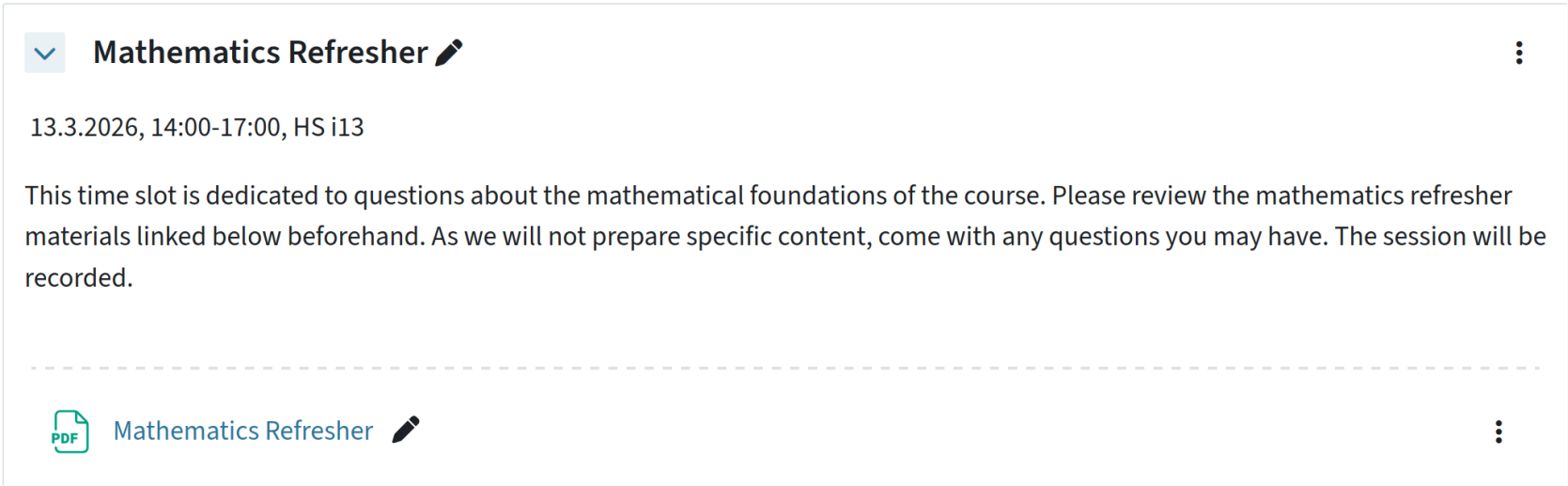
Math Refresher

"Mathematics" comes from Greek:

μάθημα: knowledge, study, learning

μαθηματικός: on the matter of that which is learned

See TeachCenter for a math refresher:



This content should be highly familiar! If you struggle with this content you are probably not ready for this course!

Math Q&A Session this Friday, 14:00-17:00 in HS i13. (optional)

VU vs. VO+UE Format

- starting this year, this course is implemented as a "Vorlesung mit integrierter Übung" (VU, lecture with integrated practicals)
- however, some curricula still follow the old format, where the content is offered as two separate courses
  - "Vorlesung" (VO, lectures)
  - "Übung" (UE, practicals)
- students registered under the old format will get two distinct grades
- the two formats are equivalent to each other
- if you are registered under two studies, one requiring VU and the other VO+UE, we will only issue the VU grade

Assessment VU Mode

- your grade is composed of
  - 3 assignment sheets (50%)
  - final exam, 75 minutes, 30 multiple choice questions (50%)
- **additionally, passing the final exam with at least half of the achievable points (30/60) is required!**
- please register for an exam slot:
  - 6 slots available on the 29th and 30th of June
  - see TUGonline
- missing the final exam needs a good excuse (e.g., medical emergency with certificate!)

Assessment VU Mode: Exam Resit

- if your failed the final exam, you have the possibility of a **single** resit on 15th of July (see TUGonline)
  - 14:00, i13
  - 15:30, i13
- also, if you missed the exam for a good reason (medical emergency) you can take the exam on 15th of July

Assessment VO+UE Mode

- for the VO+UE mode you will get two separate grades
- VO: 6 exam dates offered over the year (you need to pass only one)
- UE: 3 assignment sheets identical to those of of the VU

Assignment Sheets

- details to be discussed tomorrow
- 3 assignment sheets issued over the semester
- groups of at most 2 students
- **no LLM/AI usage allowed!** (except where explicitly stated)
- **after every submission, some groups will be invited to assignment interviews:**
  - **you need to be able to explain the task and the solution**
  - **point deduction up to 100% possible**
  - **in the case of cheating, plagiarism, LLM usage, etc. an invalid grade is issued**

Grading Scheme (VU, VO, UE)

Your grade will be determined by the percentage of achieved points, relative to the max. points.

For VU:

$$\text{total perc.} = 0.5\ \text{exam perc.} + 0.5\ \text{assignments perc.}$$

For VO and UE: the respective percentage is used

90	-	100	Sehr Gut (1)
80	-	89.9	Gut (2)
65	-	79.9	Befriedigend (3)
50	-	64.9	Genügend (4)
0	-	49.9	Nicht Genügend (5)

How to get a (good) grade?

1. **time**—spend time with the lecture materials, join the sessions
2. **routine**—drop by the sessions every week (or watch the recordings)
3. **dedication**—look for the "simplicity," the epiphany, the "aha-moment"

Contact

- live sessions on Tuesday and Wednesday
- for general questions about the course, please use the forum on TeachCenter
- in particular, please ask content related questions in the forum (others might be interested in them as well)
- for all other questions:
  - robert.peharz@tugraz.at
  - thomas.wedenig@tugraz.at

Machine Learning Paradigms

Machine Learning Paradigms

Machine Learning is understood to consists of three paradigms:

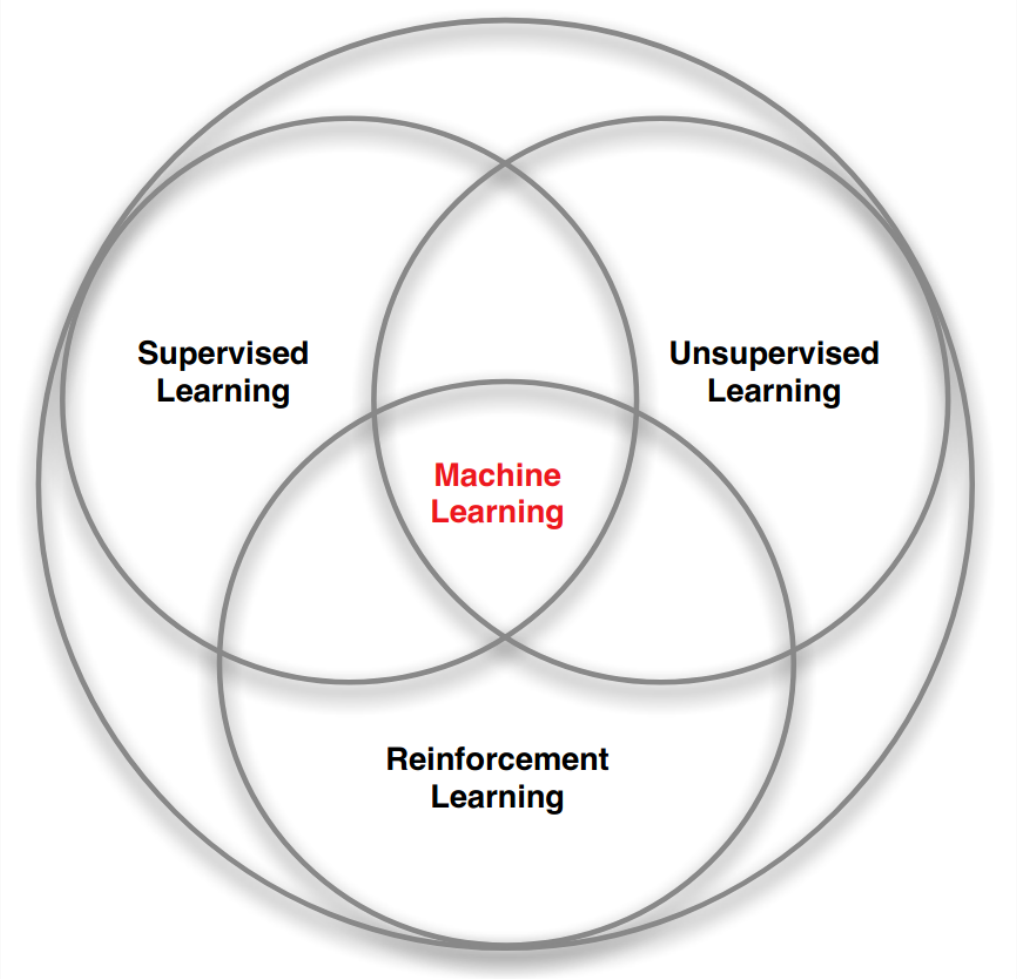


Image: David Silver

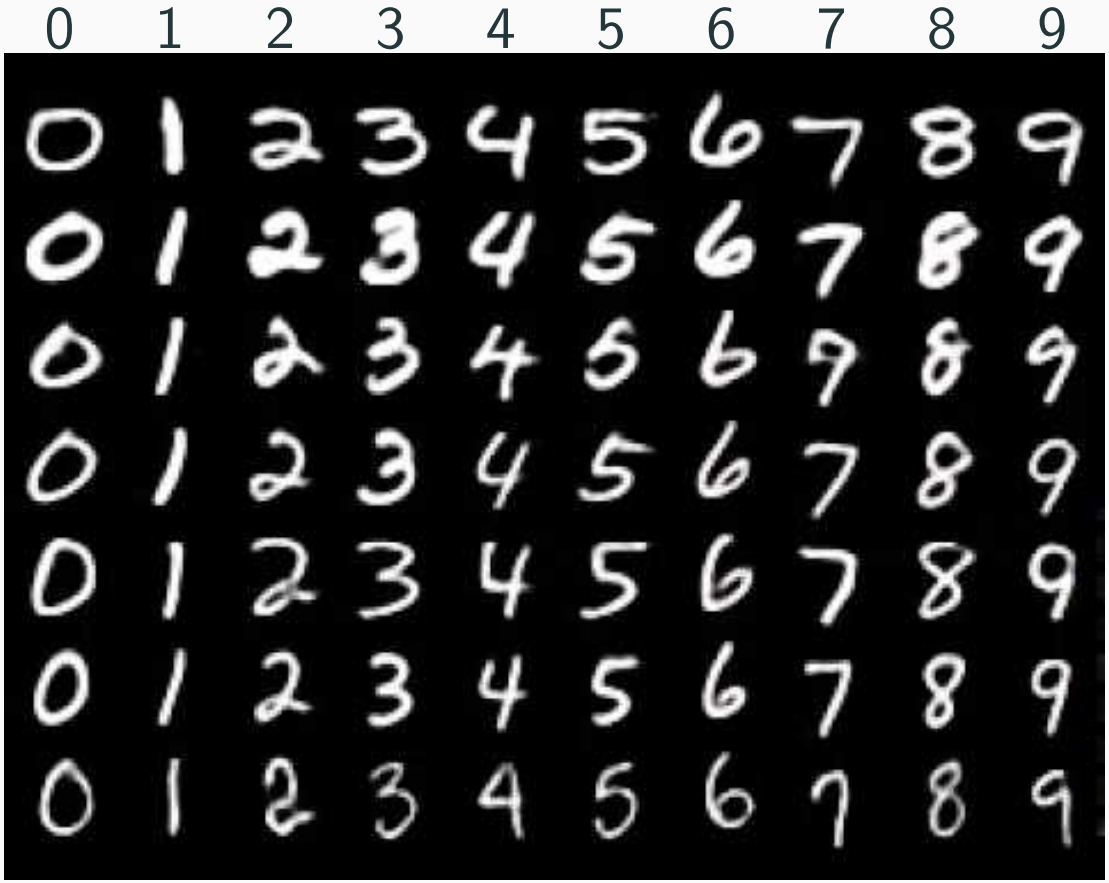
Supervised Learning

- Teacher-student situation
- Teacher gives you many examples of a concept (e.g. addition)
- For the final exam, you shall solve new (but similar) examples on your own

- **Input-output characteristics**
- **Predict target from input**

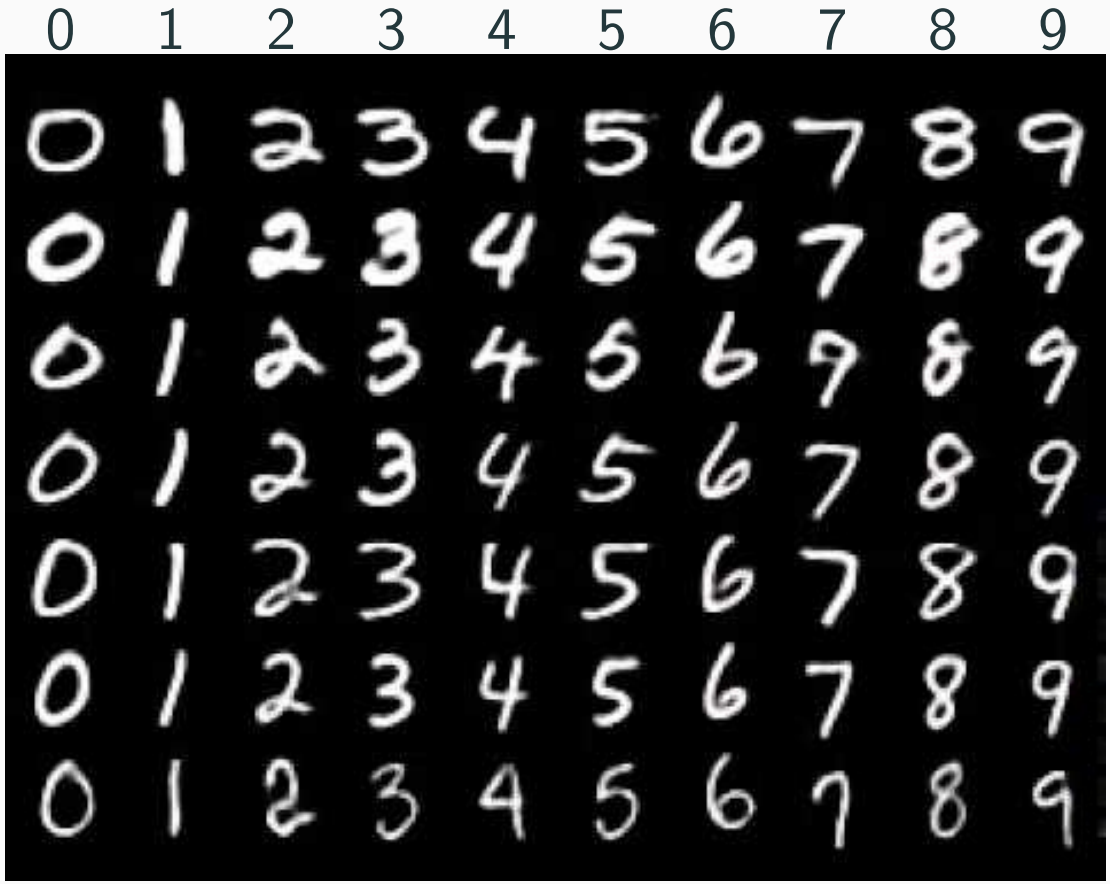
training targets (labels)

training inputs



training targets (labels)

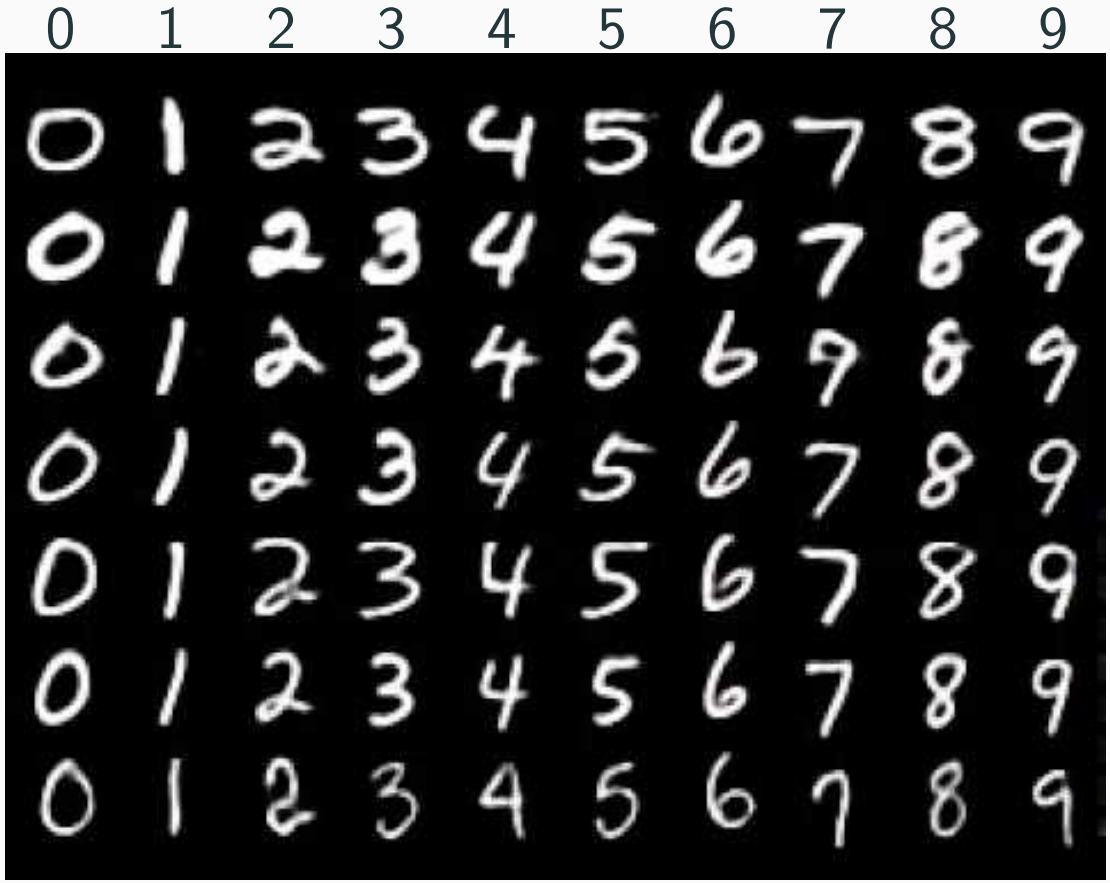
training inputs



test input (not in training data)

training targets (labels)

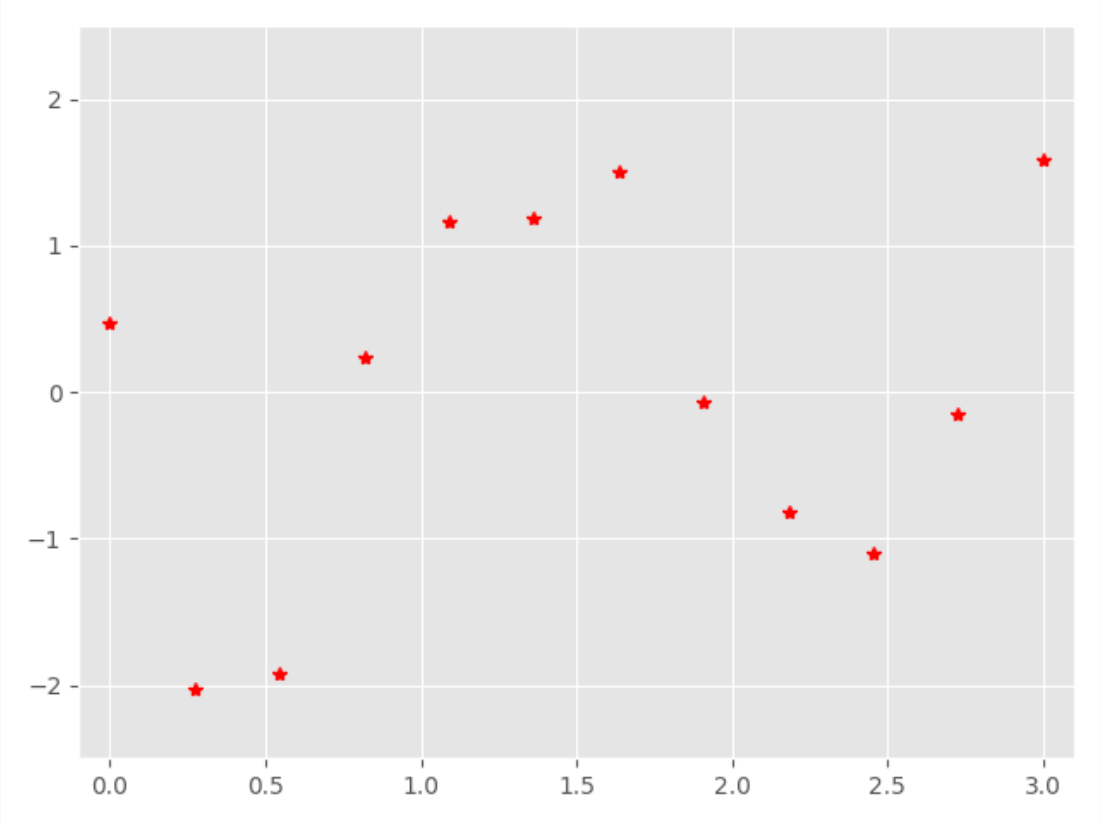
training inputs



test input (not in training data)

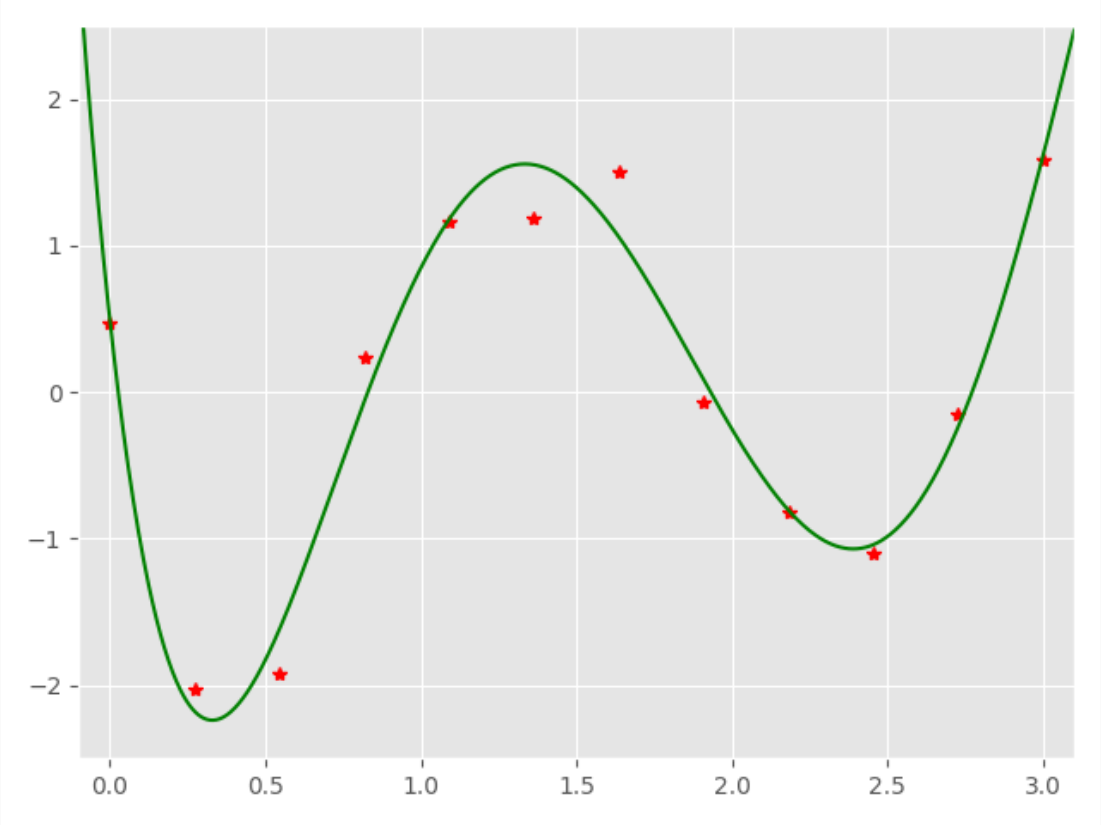
$\Rightarrow$ classifier will (hopefully) output ‘2‘

training data (independent variable x, target y)



training data (independent variable x, target y)

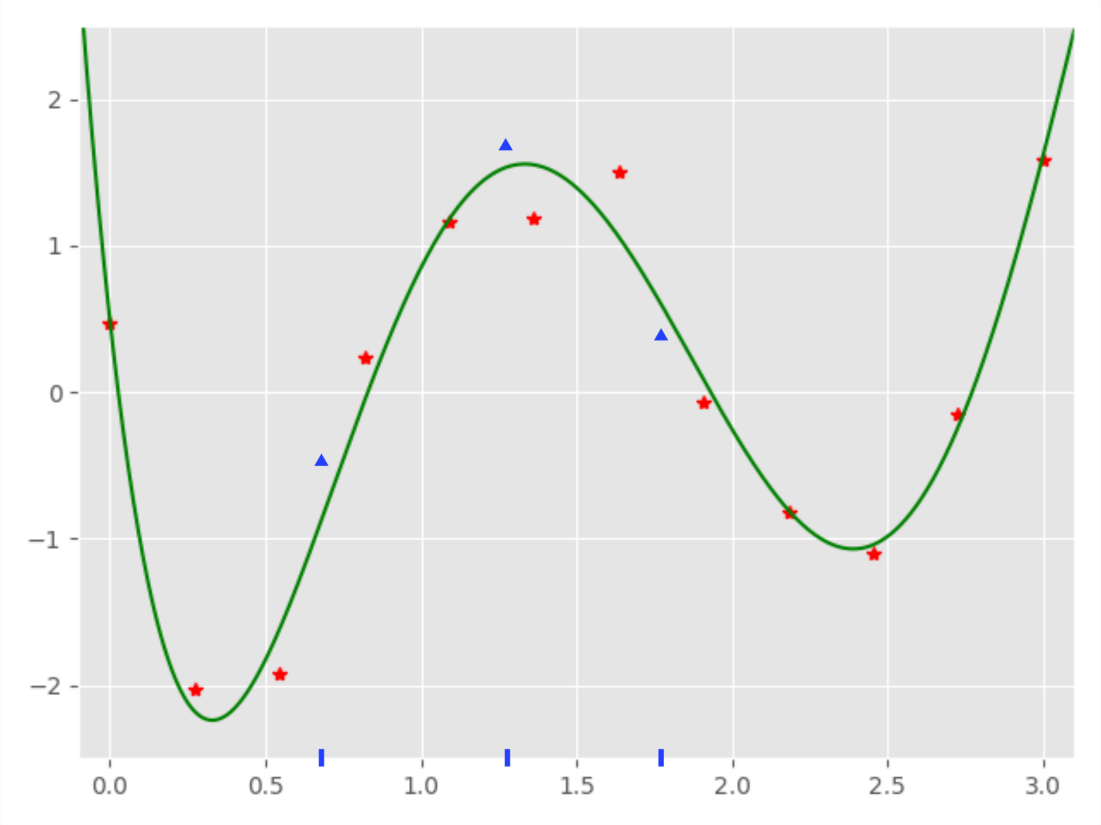
model (polynomial degree 5)



training data (independent variable x, target y)

model (polynomial degree 5)

test data prediction error on test data will (hopefully) be small



What is the key difference between **Classification** and **Regression**?

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- **Classification**: discrete, unordered targets
- **Regression**: continuous, ordered targets

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Note: there are also prediction tasks which are in between classification and regression, like Ordinal Regression, which uses discrete, but ordered targets (not covered in this course).

Machine Learning Paradigms

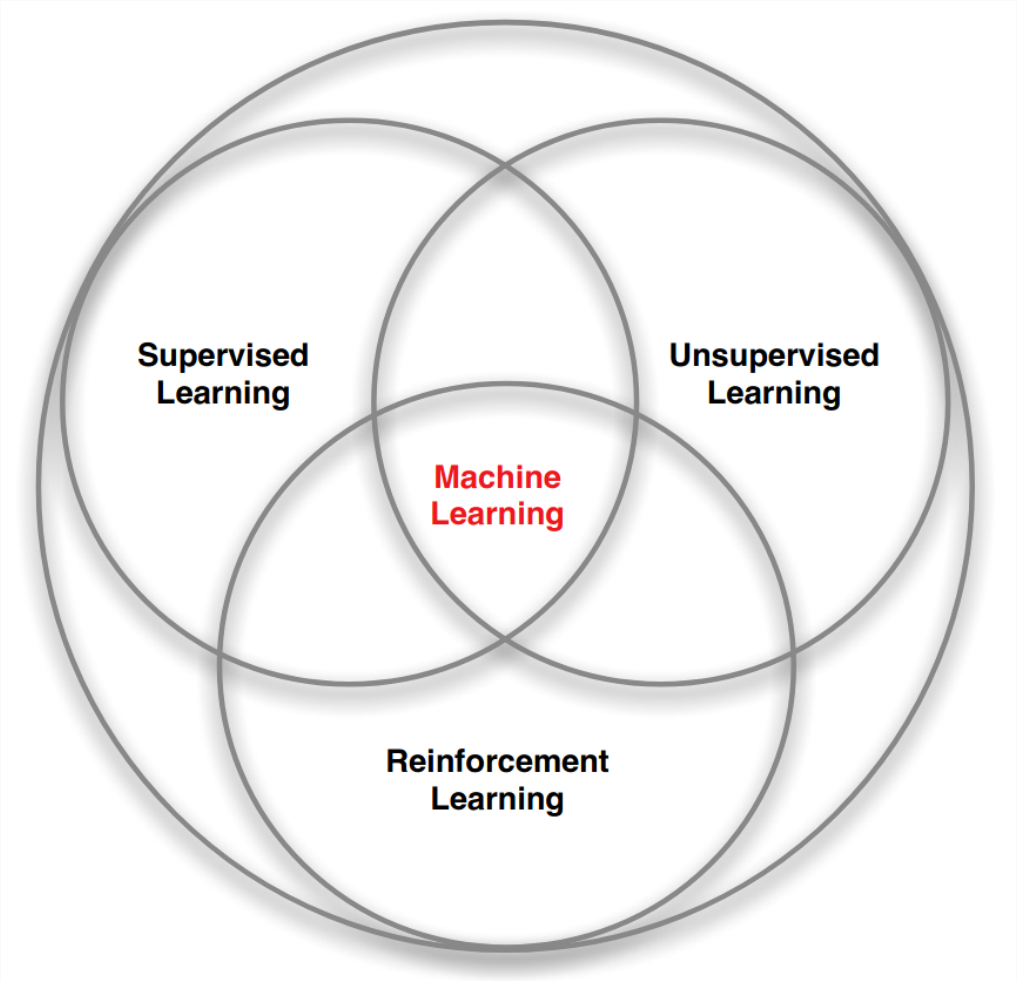

Image: David Silver

Unsupervised Learning

- You go to work by bike every day
- One day, you note a strange noise

- **No teacher**
- **Discover the world on your own**
- **Only inputs, or, outputs unknown**
- **Find interesting patterns in data**

- What are the clusters are in these data? (**Clustering**)

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- Which distribution generated these data? (**Density Estimation**, **Generative Modeling**)

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- Is this a typical example?

- What are the clusters are in these data? (**Clustering**)
- Which distribution generated these data? (**Density Estimation**, **Generative Modeling**)
- Is this a typical example?
- And this? (**Outlier Detection**, **Novelty Detection**)

Machine Learning Paradigms

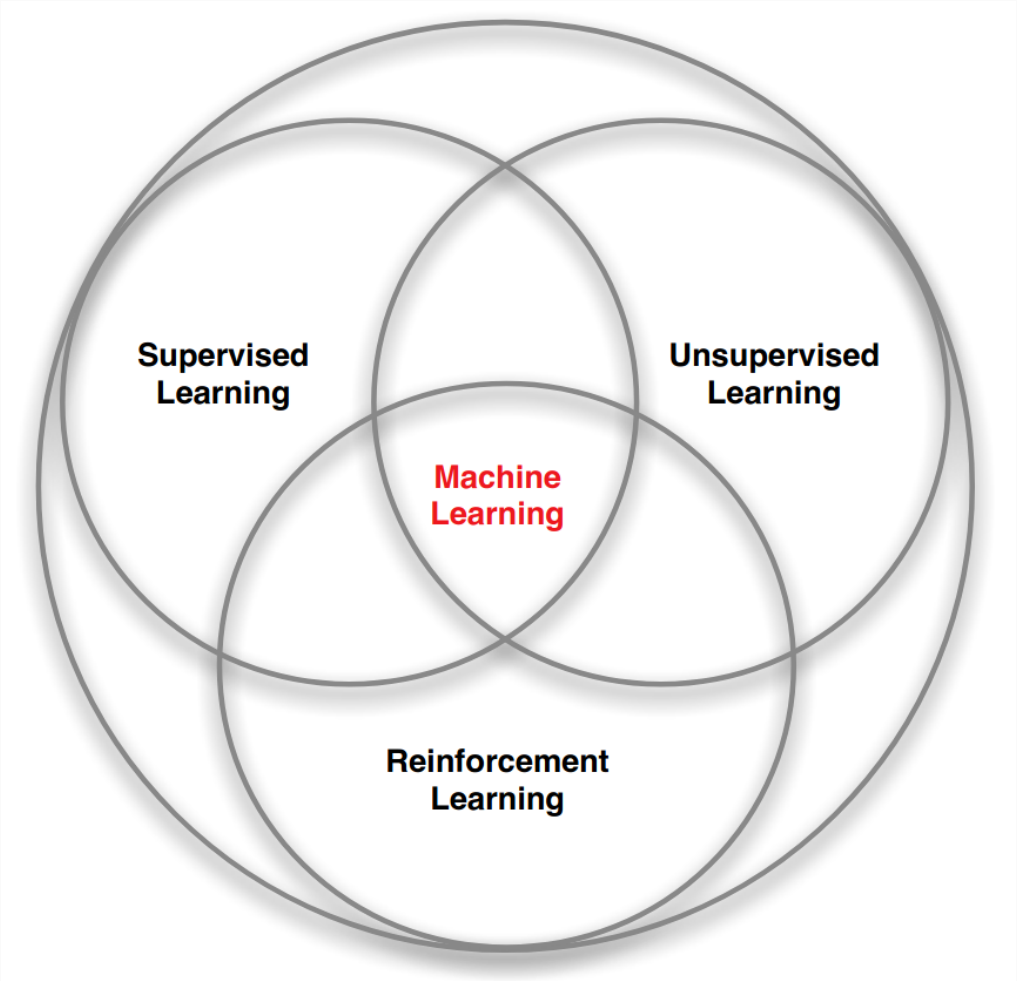

Image: David Silver

Reinforcement Learning [not covered in this course]

- You are a dog learning to sit on command
- First, you don't know what is going on...
- ...but at some point, you try to sit after command "sit!"
- And you get a tasty reward!

- **Interact with your environment**
- **Learn to make (sequential) decisions**
- **Only feedback: reward signal**

Reinforcement Learning [not covered in this course]

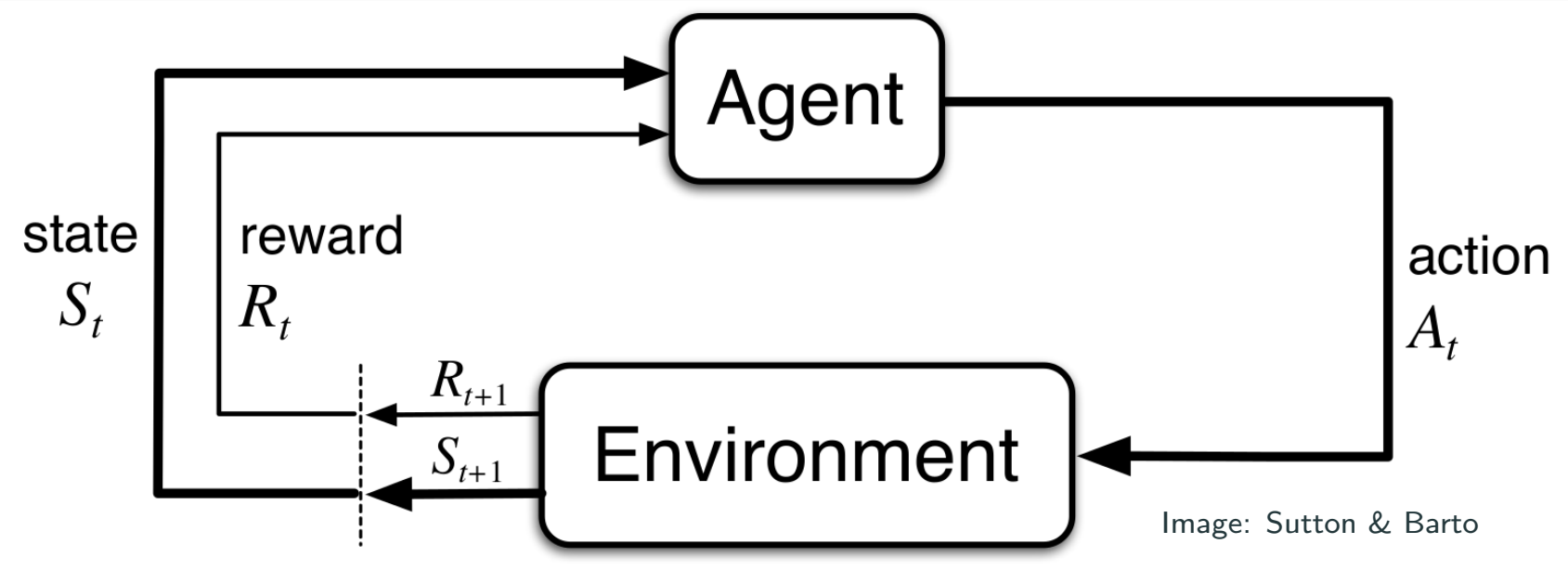

Image: Sutton & Barto

Paradigms, Paradigms, Paradigms

The traditional split of ML into **Supervised**, **Unsupervised** and **Reinforcement Learning** is perhaps somewhat outdated. Many more paradigms have been invented:

- **Semi-supervised learning** (labeled and unlabeled examples)
- **Active learning** (learn by picking examples to be labeled)
- **Bayesian optimization** (optimize an unknown function, while learning the function)
- **Imitation learning** (copy behavior)
- **Coactive learning** (user returns improved solution)
- **Self-supervised learning** (learn about the data by generating your own tasks)
- **Transfer learning** (adapt learned task to a similar task)
- **Meta learning** (learning to learn)

Course Content (might be subject to change)

1. Introduction
2. Linear regression
3. Logistic regression
4. Dimensionality reduction
5. Neural Networks 1
6. Model selection, Evaluation
7. Neural Networks 2
8. K-Nearest Neighbours, Decision Trees, Random Forests
9. Support Vector Machines, Kernel Trick
10. K-Means clustering
11. Gaussian Processes 1
12. Gaussian Processes 2
13. Gaussian Mixture Models
14. Graphical Models